

Email Spam Classifier - Step-by-Step Guide

1. Import Libraries

```
import numpy as np
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score
```

2. Load the Dataset

```
df = pd.read_csv('mail_data.csv')
print(df)
```

3. Clean the Data

```
data = df.where(pd.notnull(df))
print(data.shape)
```

4. Prepare Labels

```
data.loc[data['Category'] == 'spam', 'Category'] = 0
data.loc[data['Category'] == 'ham', 'Category'] = 1
```

```
X = data['Message']
Y = data['Category'].astype('int')
```

5. Split Data

```
X_train, X_test, Y_train, Y_test = train_test_split(X, Y, test_size=0.2, random_state=3)
```

6. Feature Extraction using TF-IDF

```
feature_extraction = TfidfVectorizer(min_df=1, stop_words='english', lowercase=True)
```

```
X_train_features = feature_extraction.fit_transform(X_train)
X_test_features = feature_extraction.transform(X_test)
```

7. Train the Model

```
model = LogisticRegression()
```

```
model.fit(X_train_features, Y_train)
```

8. Evaluate the Model

```
# On training data
```

```
prediction_on_training_data = model.predict(X_train_features)
```

```
accuracy_on_training_data = accuracy_score(Y_train, prediction_on_training_data)
```

```
print("Accuracy on training data:", accuracy_on_training_data)
```

```
# On test data
```

```
prediction_on_test_data = model.predict(X_test_features)
```

```
accuracy_on_test_data = accuracy_score(Y_test, prediction_on_test_data)
```

```
print("Accuracy on test data:", accuracy_on_test_data)
```

9. Predict on New Input

```
input_mail = ["This is the 2nd time we have tried to contact u. You have won the prize. Call now!"]
```

```
input_data_features = feature_extraction.transform(input_mail)
```

```
prediction = model.predict(input_data_features)
```

```
if prediction[0] == 1:
```

```
    print("Ham mail")
```

```
else:
```

```
    print("Spam mail")
```